

The Internet: Protocols and Layers

Lecture 2

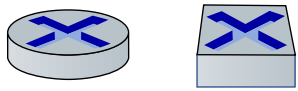


A “nuts and bolts” view of the Internet



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's “edge”



Packet switches: forward packets (chunks of data)

- *routers, switches*

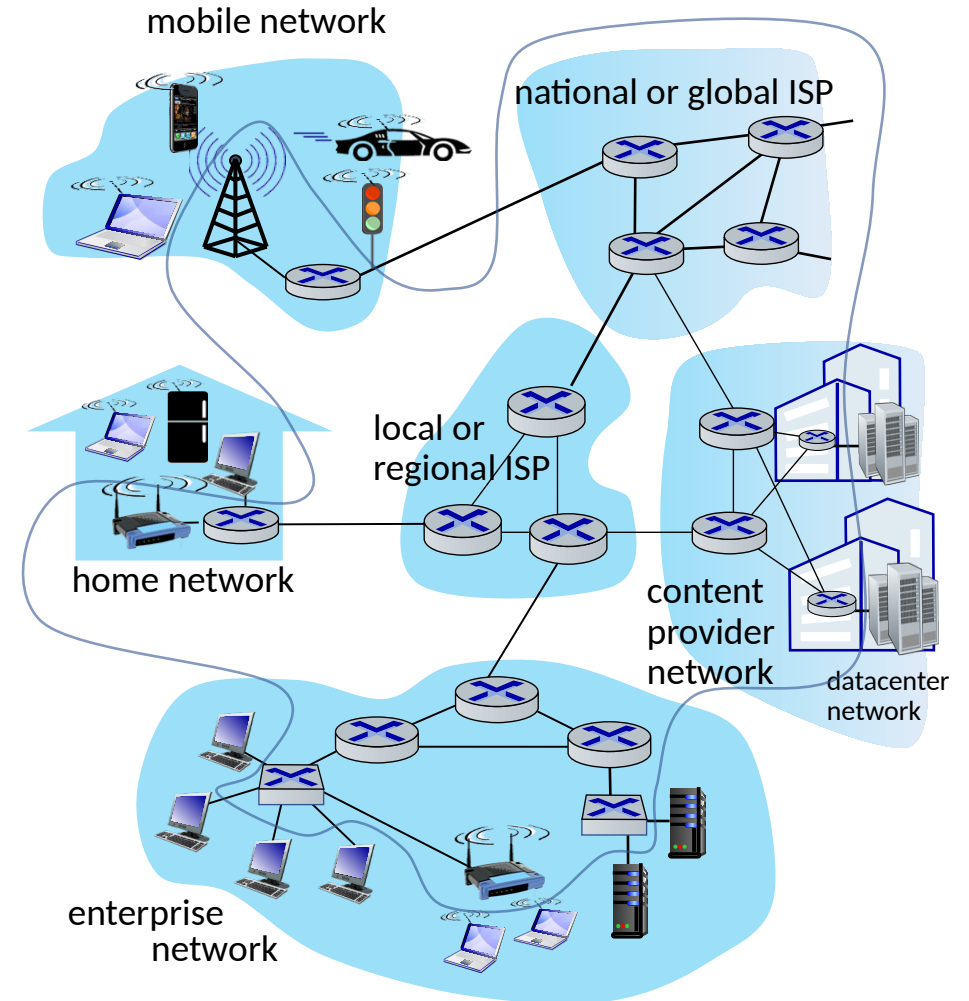


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



Various connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt:
monitor energy use



Security Camera



Slingbox: remote
control cable TV



Web-enabled toaster +
weather forecaster



AR devices

Internet phones



sensorized,
bed
mattress

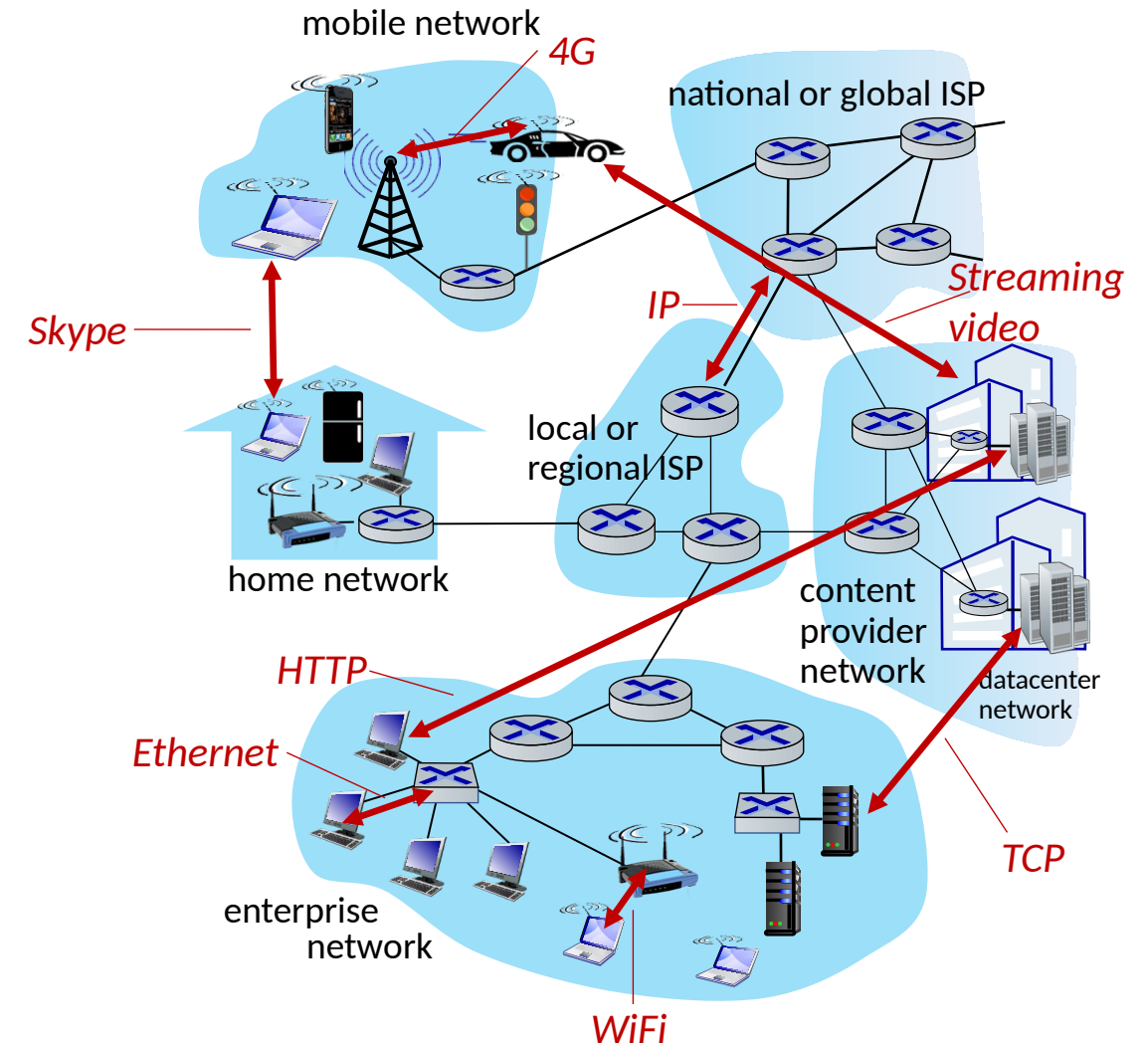


Fitbit

...and others

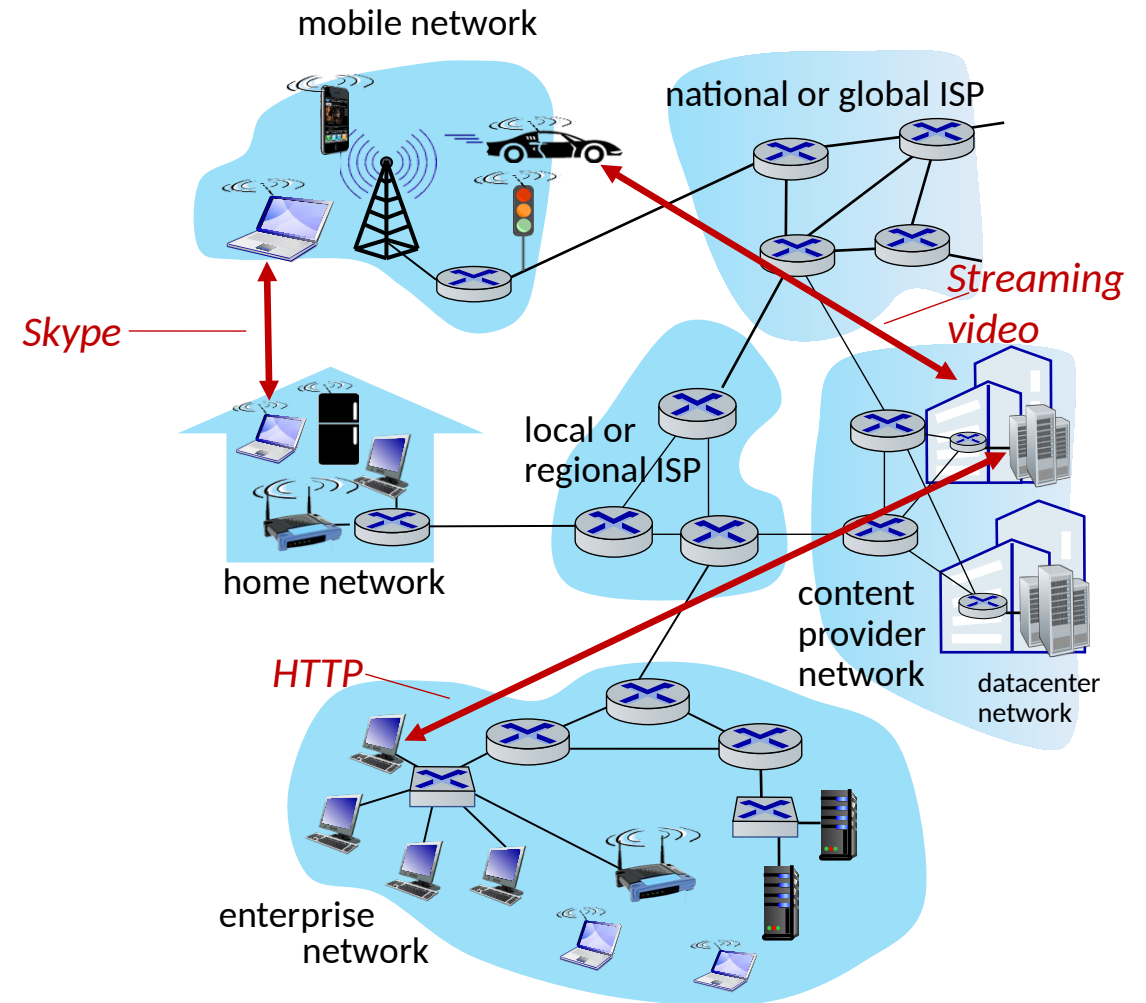
A “nuts and bolts” view of the Internet

- *Internet: “network of networks”*
 - Interconnected ISPs
- *protocols* are everywhere
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



A services view of the Internet

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



Protocols

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific messages sent

... specific actions taken
when message received,
or other events

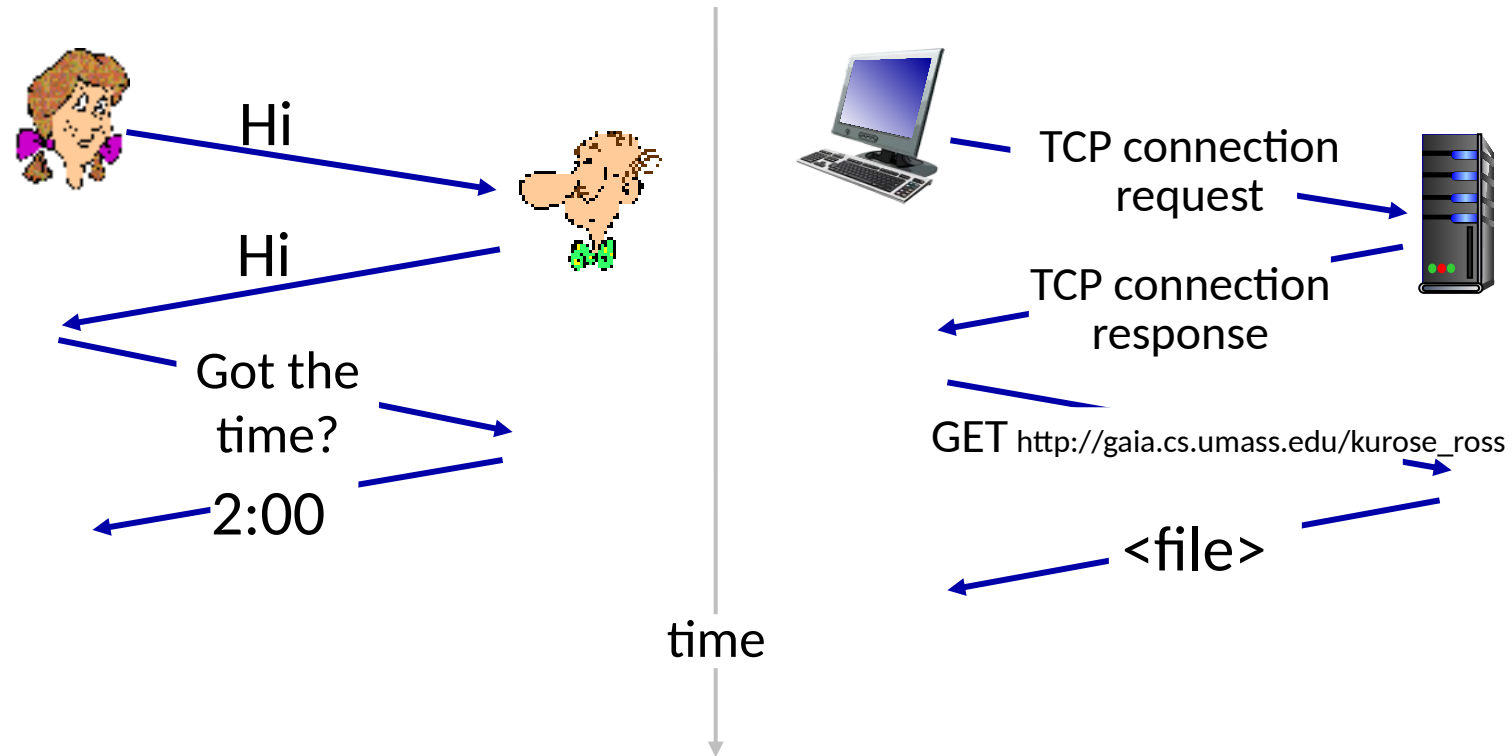
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the format, order of messages sent and received among network entities, and actions taken on msg transmission, receipt

Human and network protocols

A human protocol and a computer network protocol:



Introduction

Who Standardize Standards?

- Today's Internet standards and protocols are defined by a community of experts called **Internet Engineering Task Force (IETF)**.
- 
- The IETF logo features a stylized yellow line graph with four peaks and three valleys, set against a background of gray diamonds. Below the graph, the letters 'I E T F' are displayed in a bold, black, sans-serif font, with a registered trademark symbol (®) to the right of the 'F'.
- IETF is typically organized in **open working groups**, each one focused on specific aspects of internet.
 - Each working group produces numbered documents called **Request For Comments** (RFC) including description and definition of protocols, concepts, methods underlying a standard.
 - For example, the IPv4 protocol was defined in RFC 791 (1981).

Taming network complexity

Network architectural model

*Networks are complex,
with many “pieces”:*

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

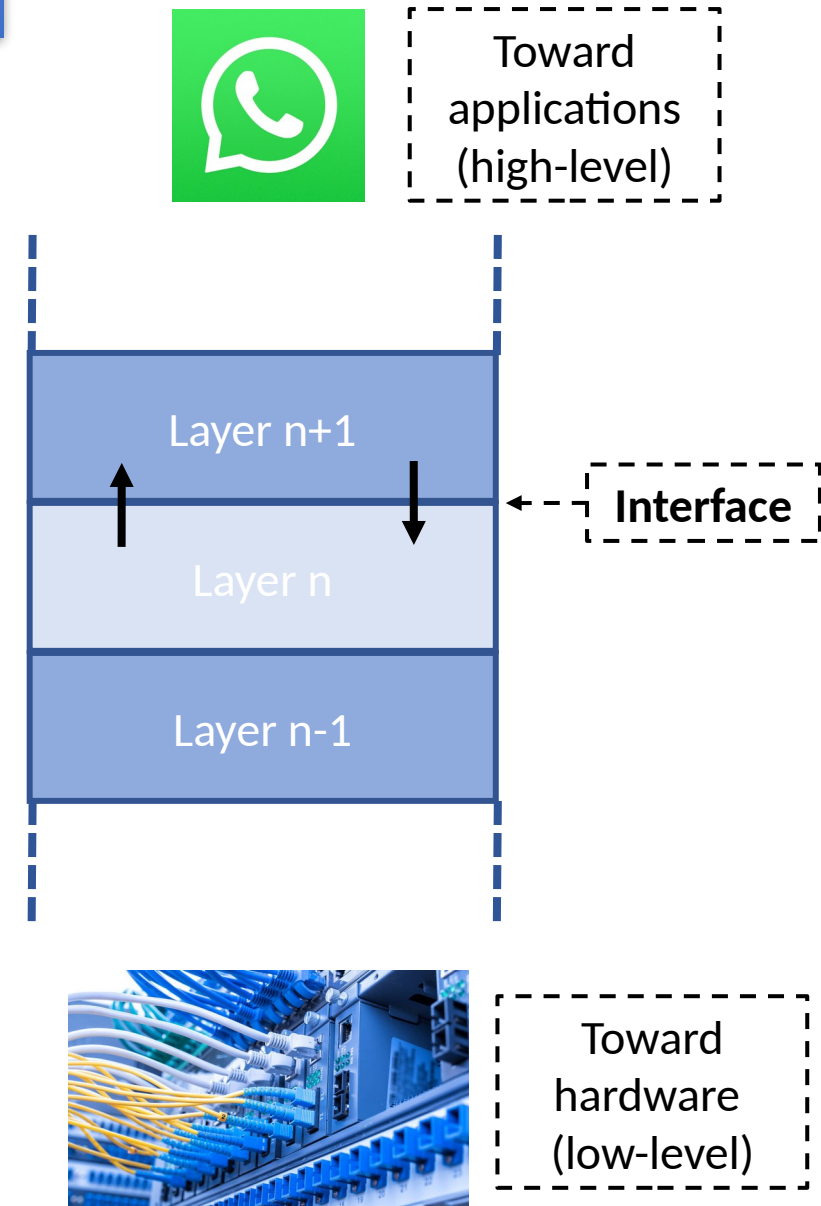
Question:

is there any hope of
organizing structure of
network?

Layering

Network model

- Architectural approach to implement communication: **layered model** (*divide et impera* in layers):
 - Each layer constitutes a **level of abstraction**.
 - Each layer **relies on services** from the lower layer and **provides services** to the upper layer.



Layering - Discussion

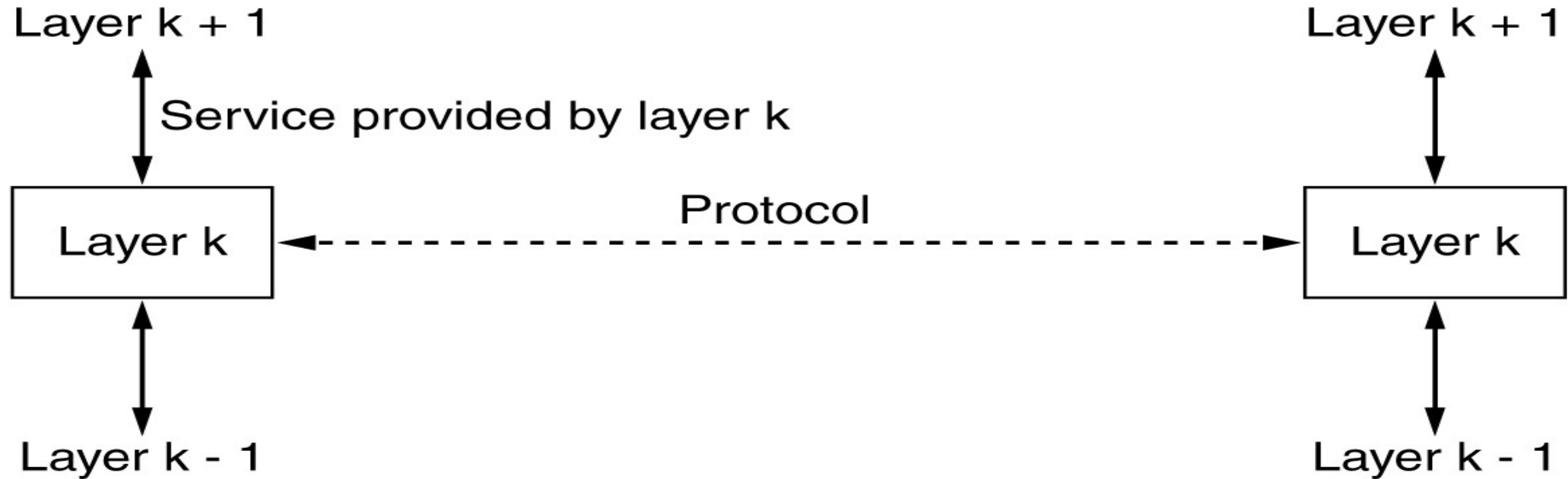
Network model

Dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
 - layered *reference model* for discussion
- modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- layering considered harmful? (efficiency)
- layering in other complex systems

Service primitives

Protocols and services

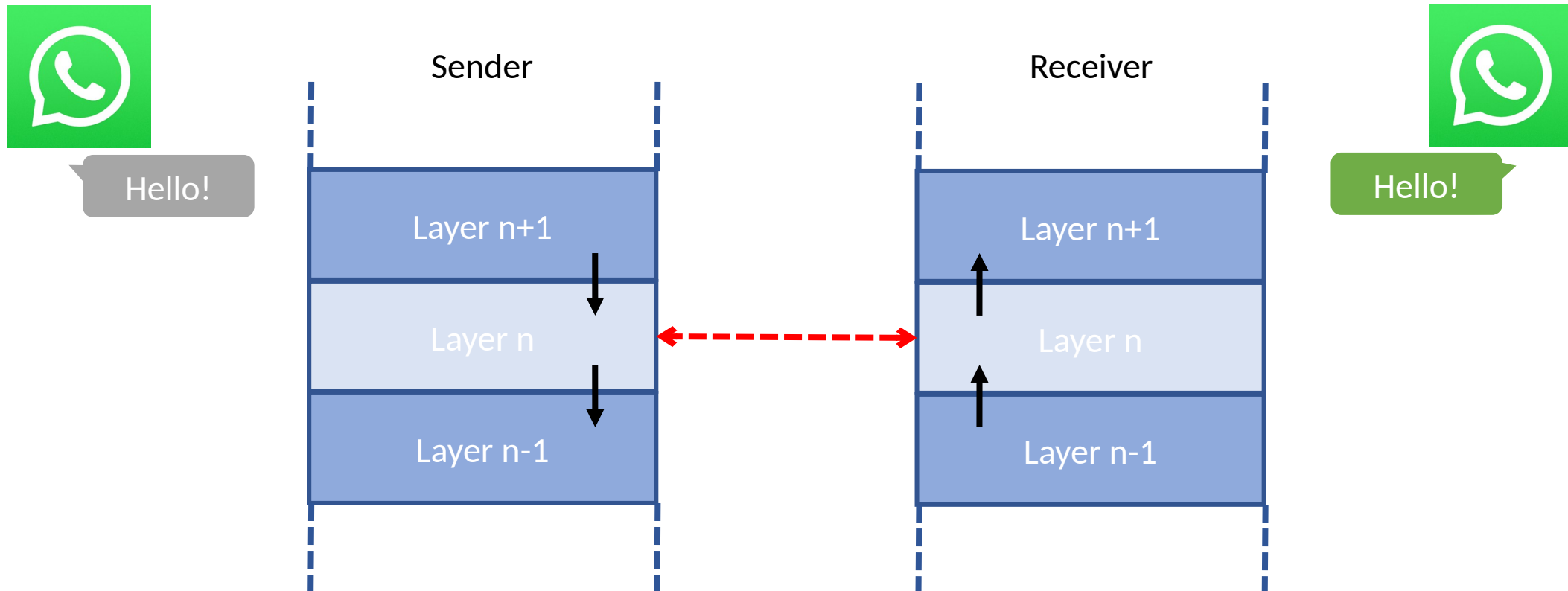


Entities use protocols in order to implement their service definitions.

Introduction

Network Models: Layered Model

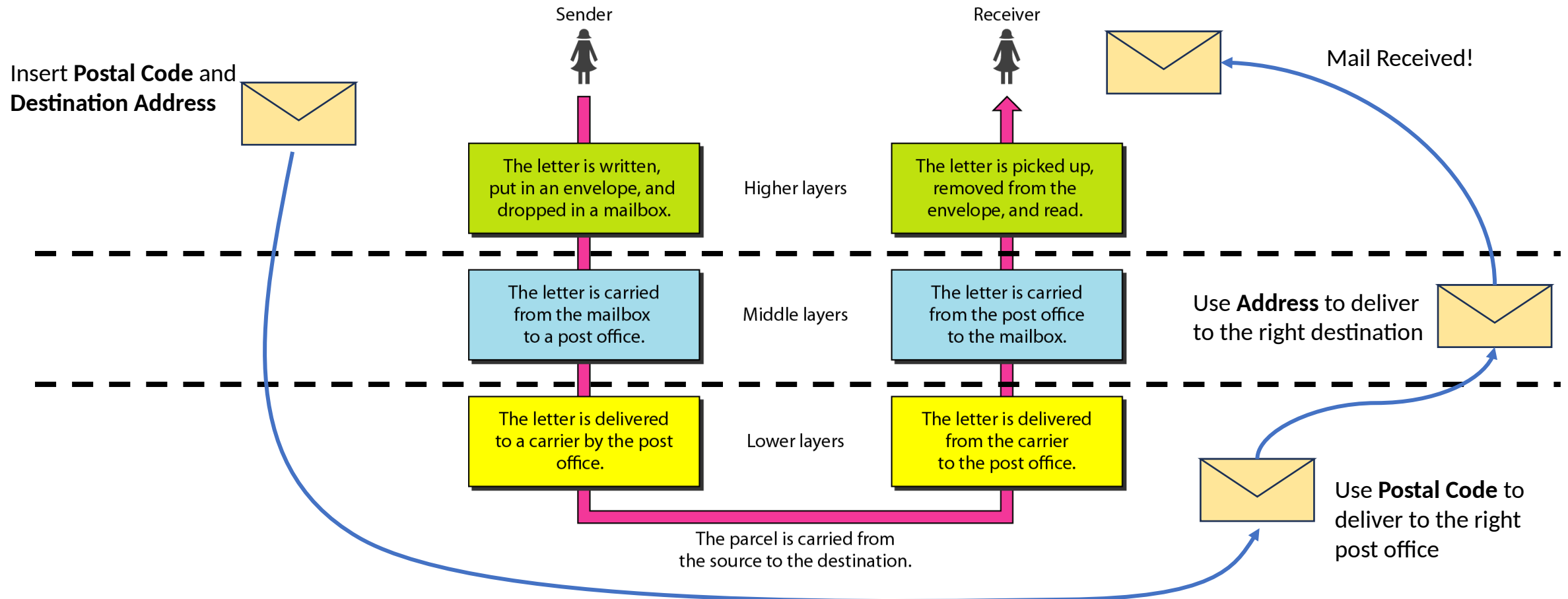
- When two devices communicate, **each layer on the sender communicates with the same layer on the receiver** by means of a specific protocol.



Introduction

Network Models: Layered Model (example)

- When two devices communicate, **each layer on the sender communicates with the same layer on the receiver** by means of a specific protocol.



Connections and reliability (1/4)

- Connection-oriented service
 - Modeled after the telephone system
 - Service user first establishes a connection, uses the connection, and then releases the connection
 - Can conduct a negotiation about the parameters to be used

Connections and reliability (2/4)

- Connectionless service
 - Modeled after the postal system
 - Packet is a message at the network layer
 - Store-and-forward switching: intermediate nodes receive a message in full before sending it on to the next node
 - Cut-through switching: transmission of a message at a node starts before it is completely received by the node
 - Datagram service: Unreliable (not acknowledged) connectionless service
- Reliability characterizes connection-oriented and connectionless services

Connections and reliability (3/4)

- Connection-oriented systems
 - Reliable message stream (sequence of pages)
 - Reliable byte stream (movie download)
 - Unreliable connection (voice over IP)
- Connectionless systems
 - Reliable message stream (electronic junk mail)
 - Reliable byte stream (text messaging)
 - Unreliable connection (database query)

Connections and reliability (4/4)

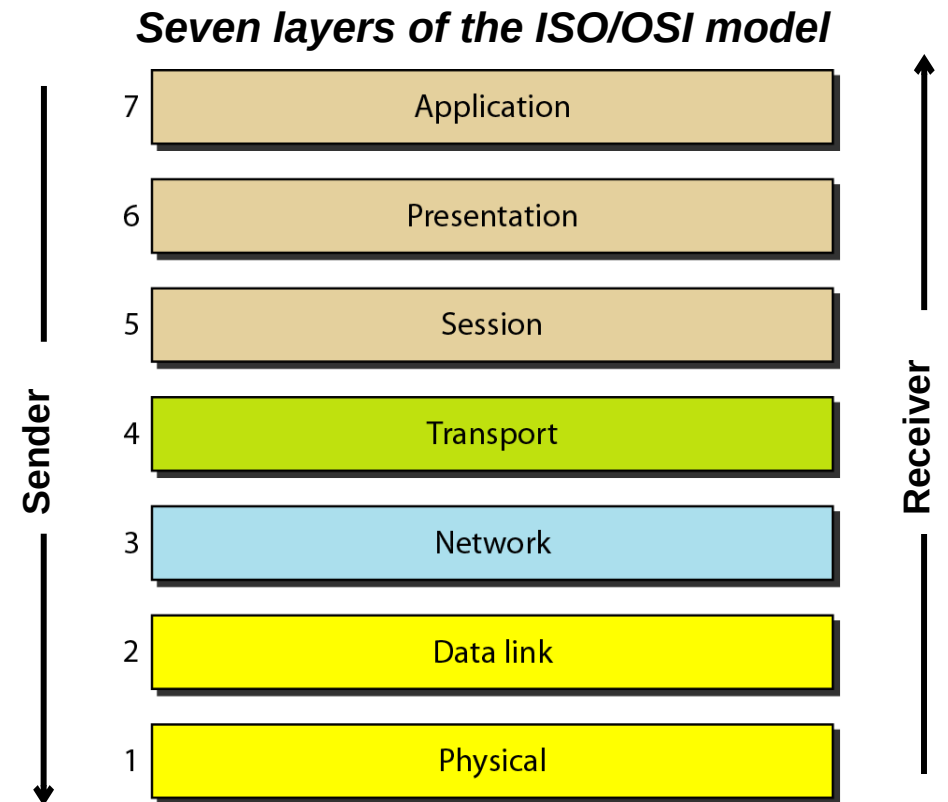
	Service	Example
Connection-oriented {	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less {	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Six common connection-oriented and connectionless services.

Introduction

Network Models: The OSI Model

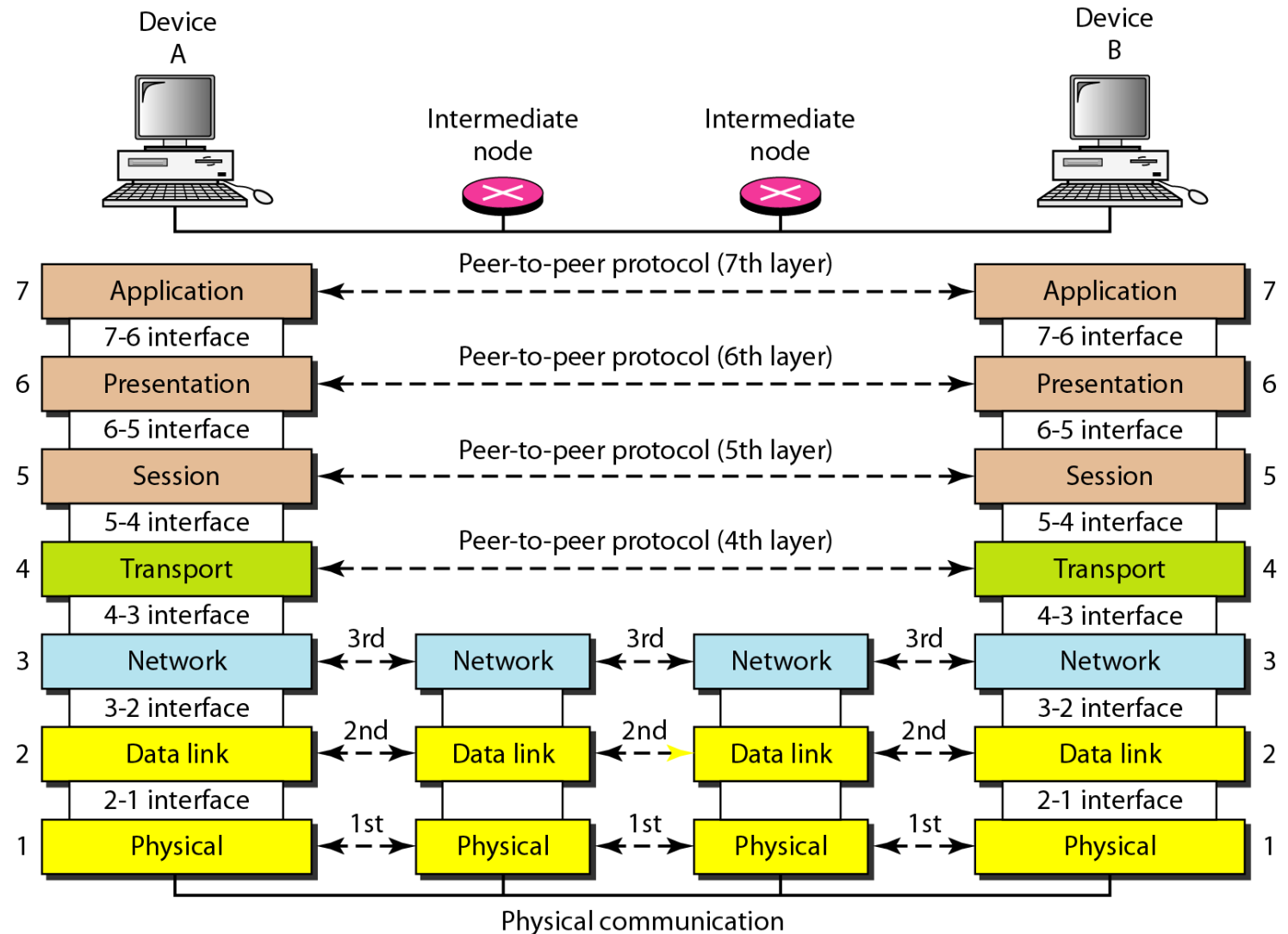
- In the '80 the **ISO** (International Standards Organization) defined a layered model for computer networks: the **OSI** (Open System Interconnection) model.
- The ISO/OSI model includes 7 layers:
 1. Physical transmission of raw bits.
 2. Data format definition (frames).
 3. Routing of messages through the net.
 4. Transmission protocols (TCP, UDP).
 5. Management of ports and sessions (continuity of data stream).
 6. Translation of messages (encoding, compression, decryption, etc.).
 7. Data use and human-computer interaction (file sharing, emails, streaming videos, etc.).



Introduction

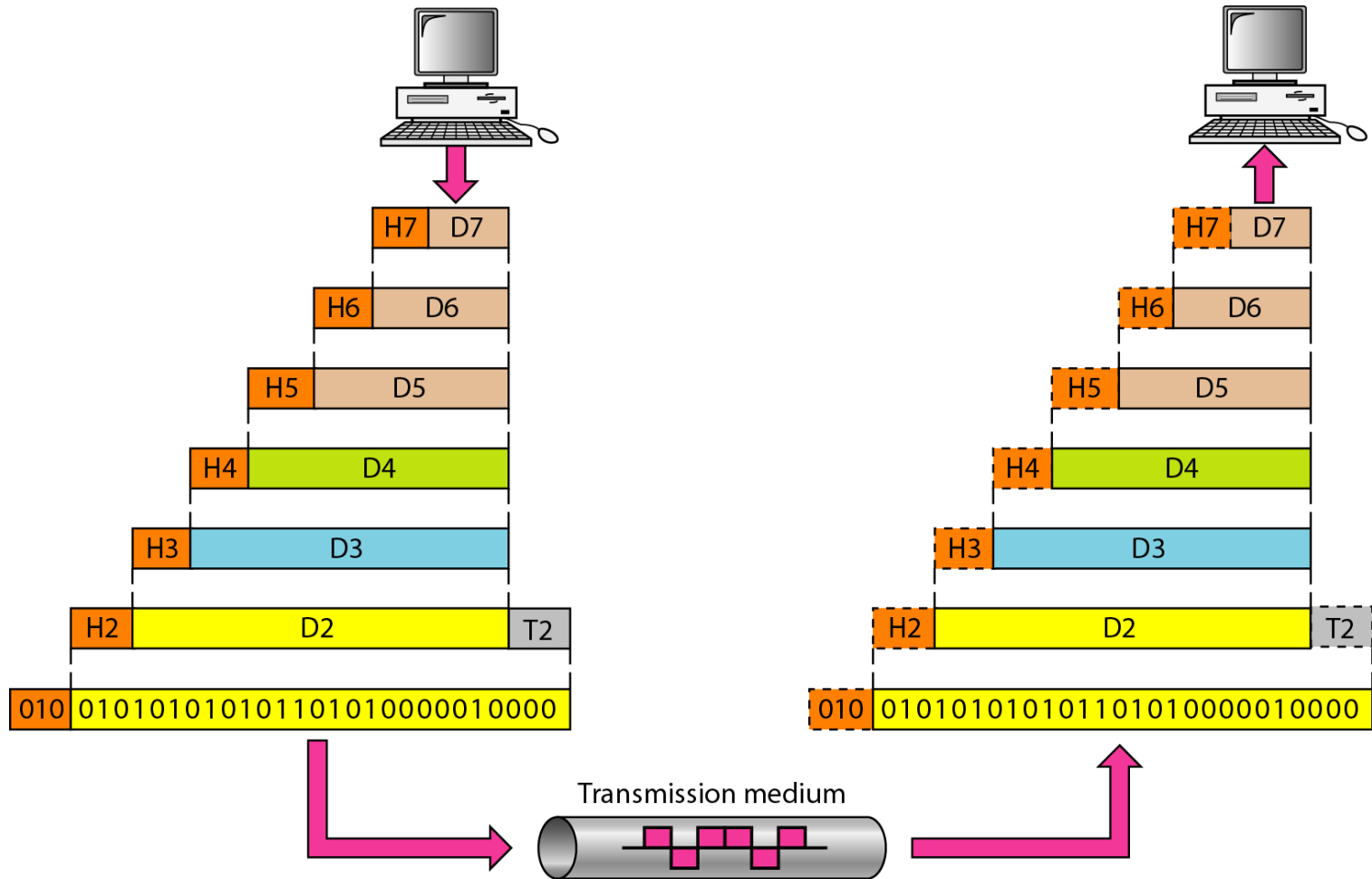
Network Models: The OSI Model

- Not always all layers are implemented.
- Typically, in intermediate nodes (switches, routers), **only 2 or 3 layers** are implemented (media layers).
- Only the **end-points** implement the **whole stack** up to the application layers.



Network Models: The OSI Model

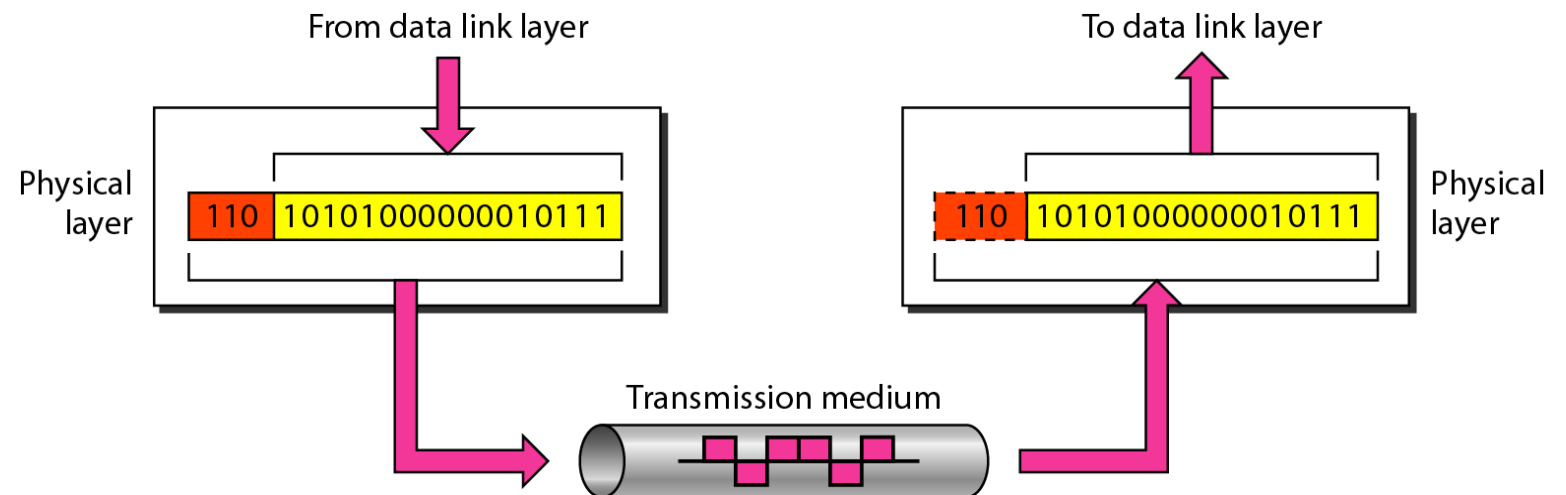
- **Decapsulation** (bottom-up): Those fields are removed and interpreted by the same layer in the receiver.



Introduction

OSI Model: 1. Physical Layer

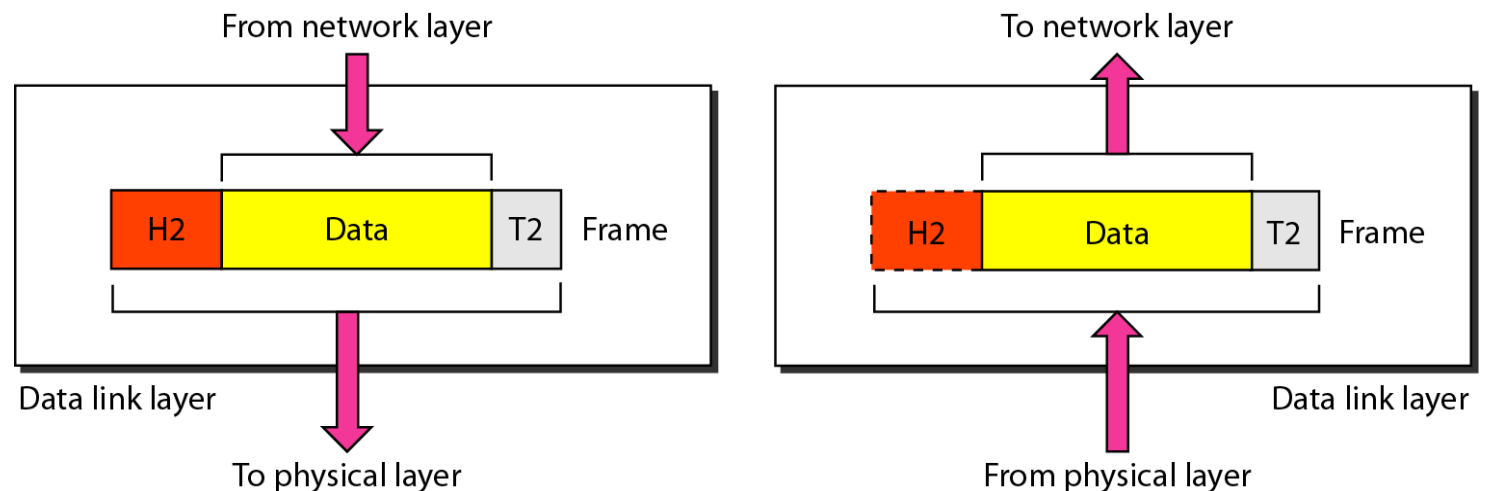
- The **physical layer** is responsible for movements of individual bits from one hop (node) to the next.
 - Physical characteristics of interfaces and media (connectors, cables, electric signals)
 - Line configuration (point-to-point or multipoint)
 - Physical topology (mesh, star, ring or bus)
 - Transmission mode (simplex, half-duplex or duplex)
 - Representation of bits
 - Data rate
 - Synchronization of bits



Introduction

OSI Model: 2. Data Link Layer

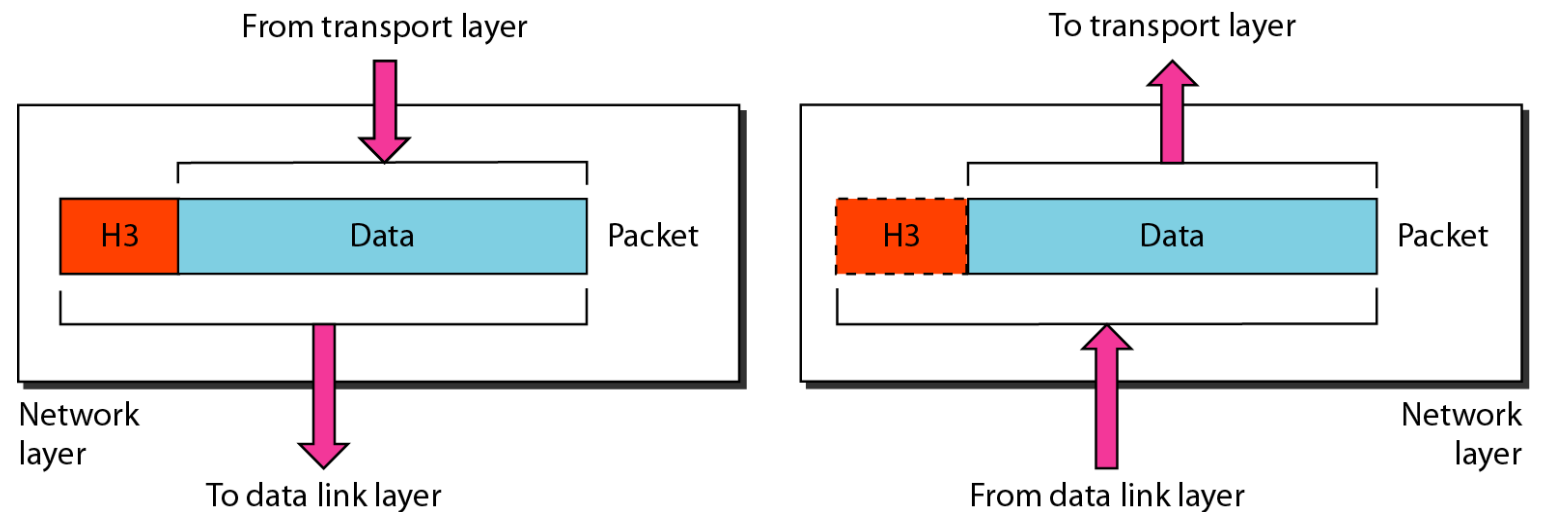
- The **data link layer** is responsible for moving frames from one hop (node) to the next (segment).
 - Works on frames (portion of data, typically few hundreds of bytes)
 - Flow and error control (frame control sequences)
 - Access control



Introduction

OSI Model: 3. Network Layer

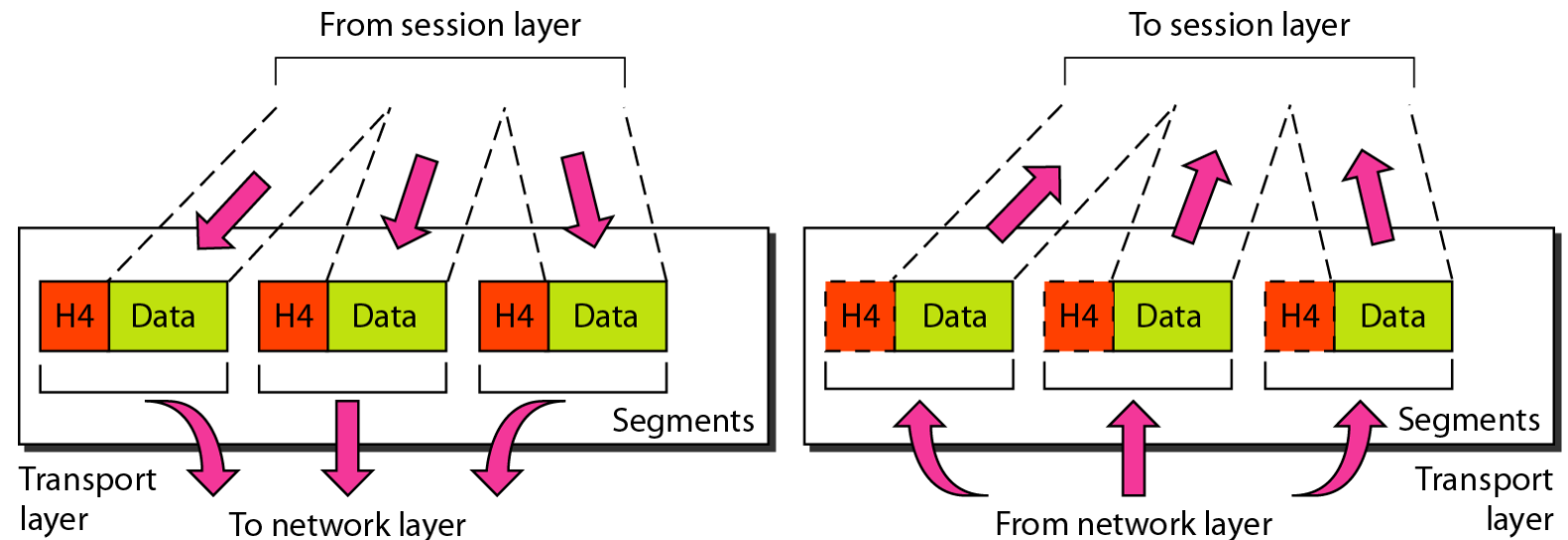
- The **network layer** is responsible for the delivery of individual packets from the source host to the destination host (path).
 - Works on packets (typically larger and more complex than frames)
 - Source-to-destination delivery: packets from the source to the destination.
 - Logical addressing
 - Routing (routing tables)



Introduction

OSI Model: 4. Transport Layer

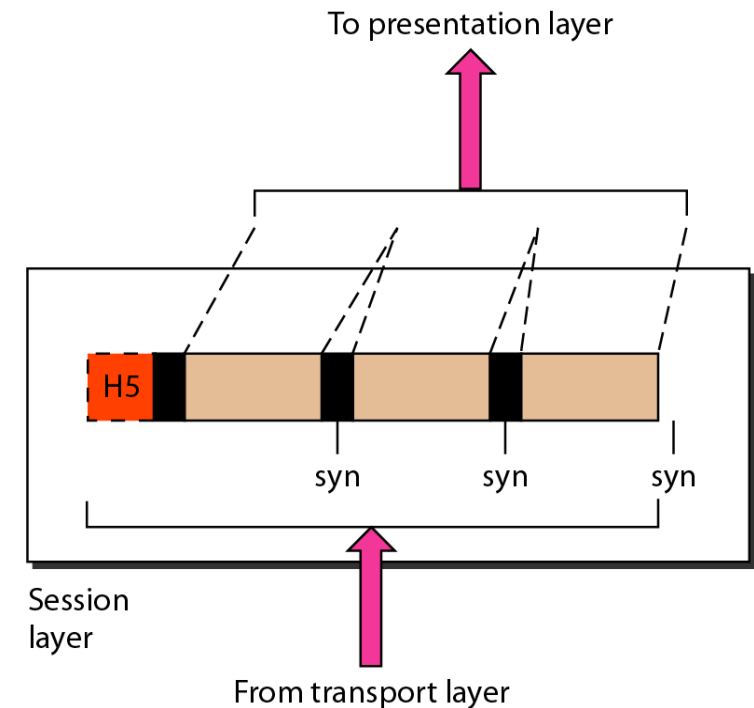
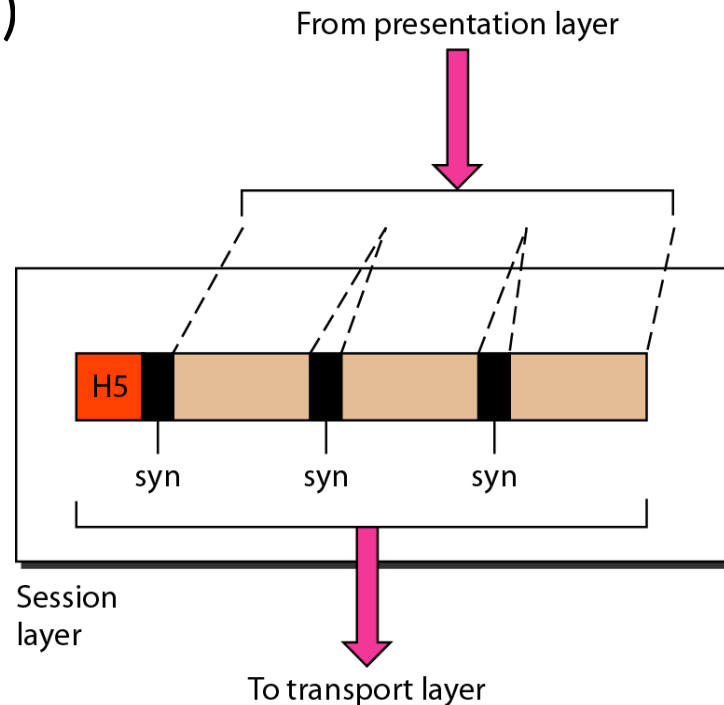
- The **transport layer** is responsible for the delivery of a message from one end to another (message is received by the right program).
 - End-to-end delivery
 - Connection control (Connection-oriented or connection-less)
 - Segmentation/reassembly (to/from packets of layer 3)
 - Port addressing
 - Flow and error control



Introduction

OSI Model: 5. Session Layer

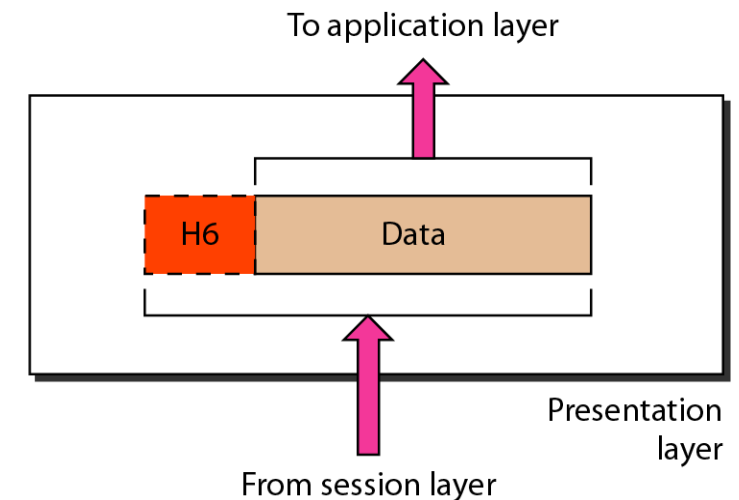
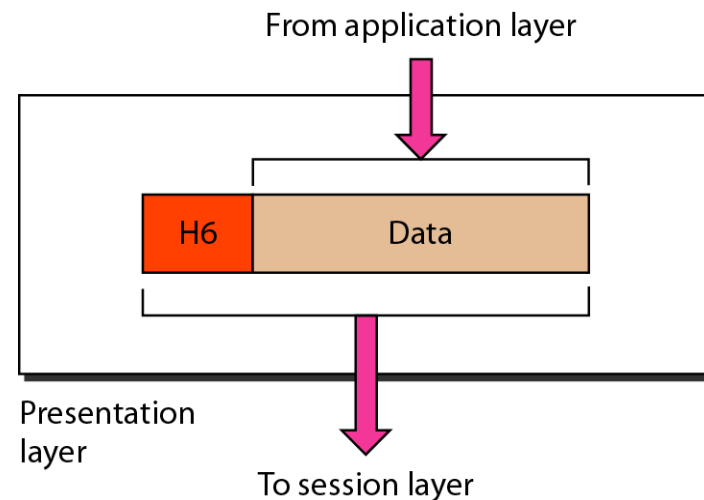
- The **session layer** is responsible for dialog control and synchronization of request/response.
 - It establishes, maintains and synchronizes the interaction between communicating system (communication session).
 - Dialog control and management
 - Synchronization (checkpoints)



Introduction

OSI Model: 6. Presentation Layer

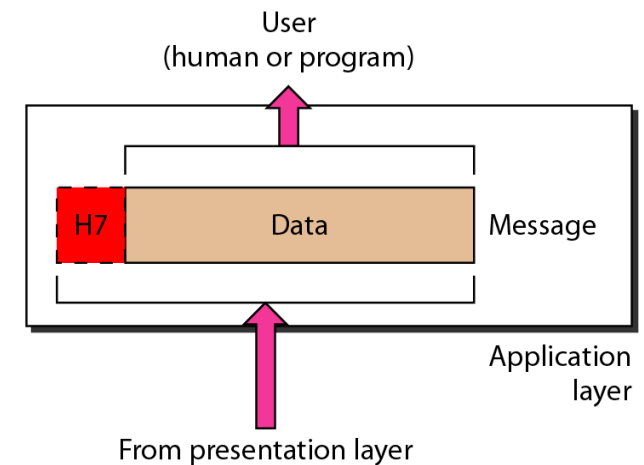
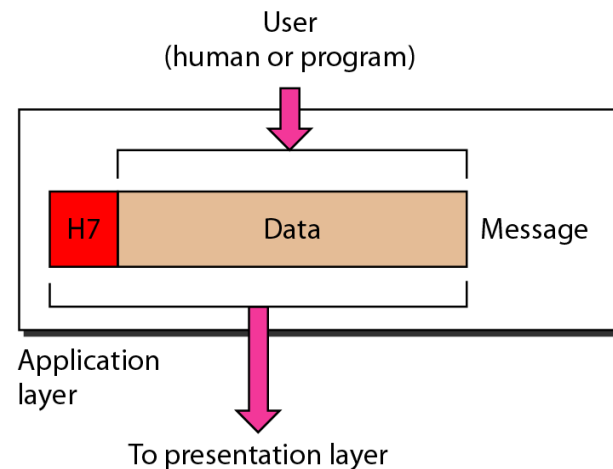
- The **presentation layer** is responsible for the representation of data, i.e., translation, compression, encryption, etc.
 - Translation (e.g., EBCDIC-coded or ASCII-coded to text)
 - Encryption and Decryption
 - Compression



Introduction

OSI Model: 7. Application Layer

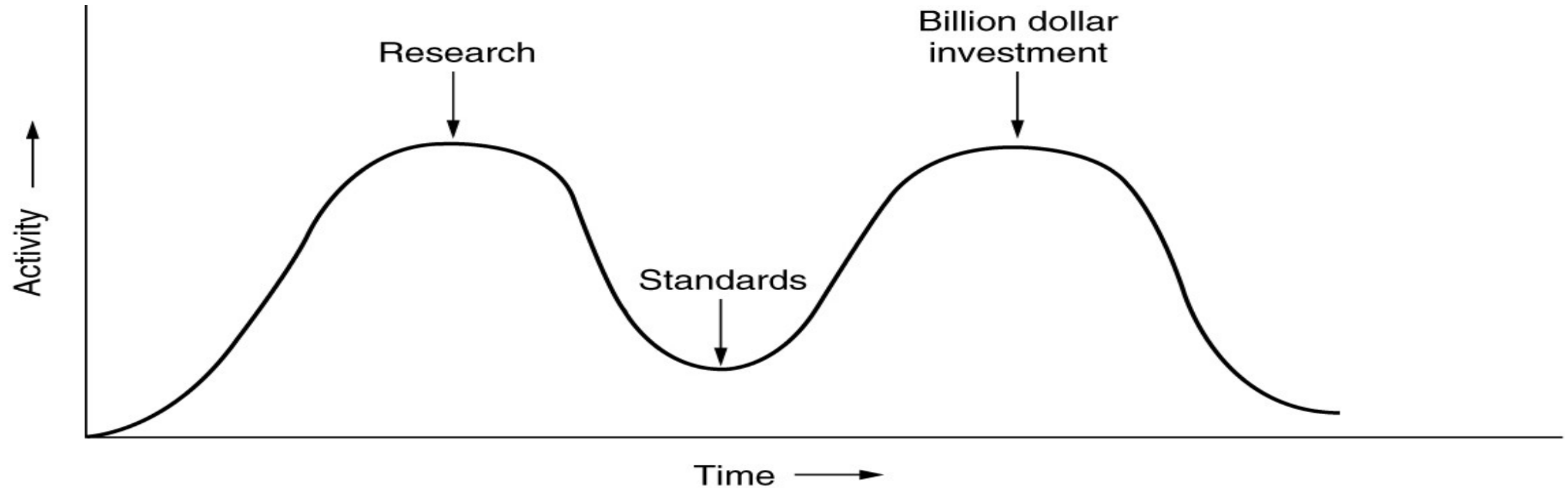
- The **application layer** is responsible for providing services to the user.
 - Network virtual terminal (Remote log-in)
 - File transfer and access
 - Mail services
 - Accessing the World Wide Web
 - ...



Critique of the OSI model (1/2)

- Bad timing
 - Competing TCP/IP protocols were already in widespread use
- Bad design
 - Both the model and the protocols are flawed
- Bad implementations
 - Initial implementations were huge, unwieldy, and slow
- Bad politics
 - Widely thought to be the creature of the European telecommunication ministries, the European Community, and later the U.S. Government

Critique of the OSI model (2/2)



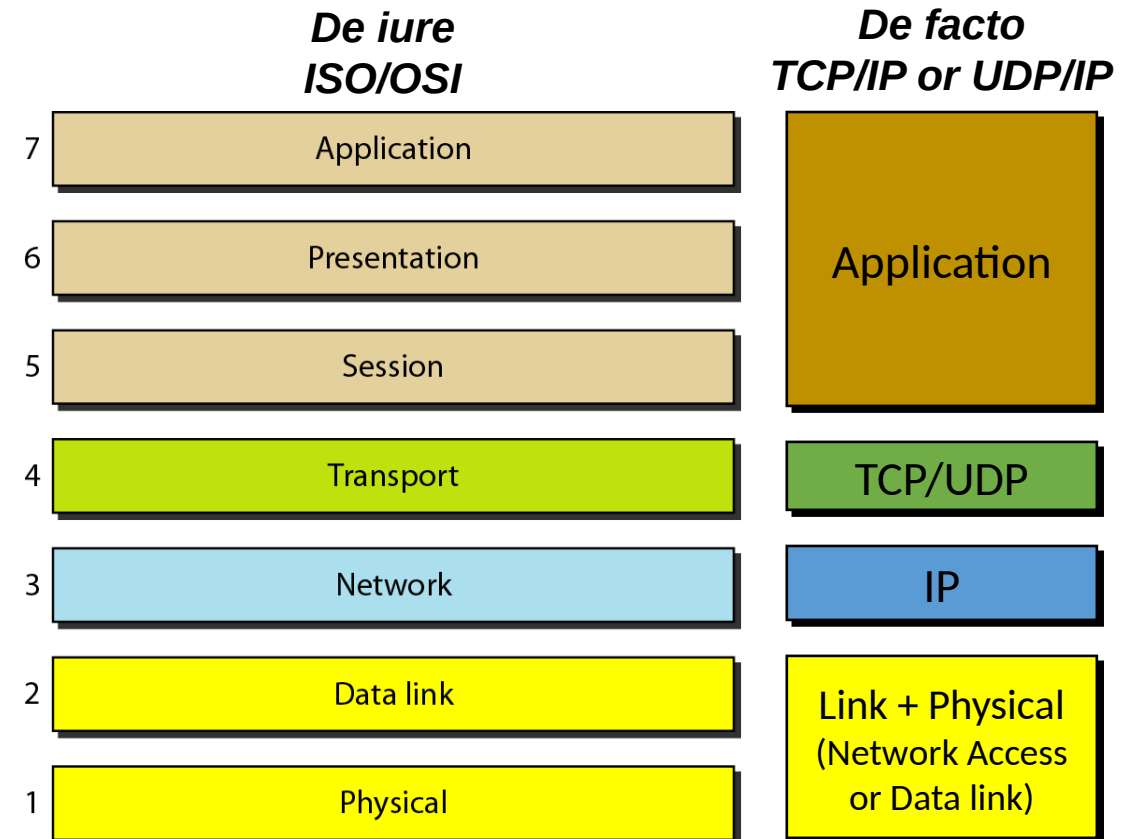
To prevent bad timing, it is essential that the standards be written in the trough in between the tops of the waves known as the two “elephants.”

Critique of the TCP/IP model (1/2)

- Model does not clearly distinguish the concepts of services, interfaces, and protocols
- Model is not at all general
 - Poorly suited to describing any other protocol stack
- The link layer is not really a layer at all in the normal sense of the term
- Model does not distinguish between the physical and data link layers
- Other protocol implementations were distributed free

TCP/IP vs OSI

- The ISO/OSI model is *de iure* the standard stack for computer networks, but it is quite complex and detailed.
- The TCP/IP and UDP/IP (aka Internet Protocol Suite) are the set of communication protocols **actually used** on internet and local networks.
 - TCP: Transmission Control Protocol.
 - UDP: User Datagram Protocol.
 - IP: Internet Protocol.
- The TCP/IP and UDP/IP are *de facto* the standard stack for internet communication.



Other network models

- There are different protocols for device communication, which can be used depending on the situation, that are still relying on the layered model.

